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Started	Tuesday, 13 October 2026, 4:16 PM
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Duration	3 mins 31 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**
Correct
Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O👎
c. O(1)
d. O(log n) ✓

Question **2**
Correct
Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(log n)
c. O(1) ✓
d. O👎

Question **3**
Correct
Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear
b. Both (1) & (3)
c. Hash
d. Binary ✓

Question **4**
Correct
Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search continues from the beginning
c. The search continues in the left or right sublist ✓
d. The search stops

Question **5**
Correct
Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Sequential search
- b. Binary search ✓
- c. Linear search
- d. Hash search

Question **6**
Correct
Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search does not require dividing the list
- b. Binary search can find multiple instances of the target element
- c. Binary search has a lower time complexity on large, sorted lists ✓
- d. Binary search works on unsorted lists

Question **7**
Correct
Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. There must be a mechanism to access middle element directly
- b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- c. Must use a sorted array
- d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **8**
Correct
Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 100
- c. 1000
- d. 10 ✓

Question **9**
Correct
Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It compares elements sequentially
- b. It can be applied only if the list is sorted ✓
- c. It works on unsorted lists
- d. It always starts from the beginning of the list

Question **10**
Incorrect
Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Divide the list into two equal parts
- b. Sort the list ✗
- c. Compare the target element with the middle element in the list
- d. Compare the target element with the first element in the list

Question **11**
Correct
Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search backtracks to find duplicate elements
- b. The search continues until the end of the list
- c. The search stops immediately ✓
- d. The search starts over from the beginning

Question **12**
Correct
Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is large and sorted
- b. When the list is small and unsorted ✓
- c. When the list is small and sorted
- d. When the list is large and unsorted

Question **13**
Correct
Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Finding
- b. Searching ✓
- c. Mining
- d. Discovering

Question **14**
Correct
Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Inserting elements into a list
- c. Sorting elements in a list
- d. Deleting elements from a list

Question **15**
Correct
Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search is more efficient than binary search on large lists.
- b. Linear search can be applied to both sorted and unsorted lists. ✓
- c. Linear search requires the list to be sorted.
- d. Linear search divides the list into halves.